CHAPTER

13

Interim Restorations

When confronted with extensive, complex treatment plans, judicious sequencing is vital to both the operator and the patient. The interim restorations or the intermediate restorative materials enhance patient motivation, which leads to successful completion of the chosen treatment plan. Currently, the term 'transitional' or 'prototype restoration' is preferred.

Interim restorative materials are temporary or semi-permanent materials, which are used until a permanent restoration is inserted. The selection of these materials depends on the configuration of the preparation. The eventual restoration, which is planned and also the time period required for the permanent restorations.

The interim material is chosen, which can be easily placed and easily removed without injuring the actual shape of the preparation.

INTERIM RESTORATIVE MATERIALS

Rationale

a. In routine, the operative treatment may not be completed in one visit. In such cases, it is necessary to fill the prepared cavity with an intermediate restorative material during the inter-appointment phase. Intermediate restorations allow temporary functioning of the tooth, maintain intra-arch relationships, prevent supraeruption and also provide esthetics. Additionally, it allows the pulp dentin organ and periodontium to recover before placement of permanent restoration.

b. In subjects, where multiple teeth are affected by acute caries, the infected tooth structure is removed and the defects restored with a temporary material. With this technique, most of the infecting organisms are removed and further spread of caries is arrested. Further, the interval between initial treatment and the permanent restoration provides time to assess the patients' compliance with oral hygiene instructions and the progress/arrest of caries activity.

c. Intermediate restorative materials are helpful in teeth with questionable pulpal prognosis. They limit the progress of demineralization in dentin and allow the underlying pulp to heal. These are also used in direct and indirect pulp capping procedures. This allows time for the underlying periodontal tissues to heal as well as form a reparative dentin barrier.

Ideal Requirements

The requisites of an ideal material for interim restoration are:

- Easy and rapid mixing
- Easy to place in cavity
- Rapid set to ensure stability
- Good strength, especially when used in larger cavities and for full crowns
- High abrasion resistance
- Low oral solubility
- Dimensional stability (coefficient of thermal expansion should match that of tooth)
- Good marginal sealing
- Easy to remove to facilitate placement of permanent restorations
- Non-irritant to pulp and other oral soft tissues
- Tasteless and odorless
- Acceptable color, especially for use in anterior teeth
- Readily available

Classification

The interim restorative materials are classified as follows:

A. For intracoronal preparation
   I. Gutta-percha
   II. Dental cements
      a. Zinc oxide eugenol cement and its modified forms
      b. Cavit
      c. Zinc phosphate cement and its modified forms
      d. Polycarboxylate cement
      e. Calcium hydroxide

B. For extracoronal preparation
  I. Prefabricated crowns:
    a. Celluloid crowns
    b. Polycarbonate crowns
    c. Stainless steel crowns
    d. Aluminum crowns
  II. Indirect acrylic restorations

A. FOR INTRACORONAL PREPARATION

I. Gutta-Percha

Gutta-percha, popularly known as Hill's stopping, is supplied in three forms for dental use (Fig. 13.1):

- Low heat (softens below 200°F)
- Medium heat (softens within the range of 200°F–210°F)
- High heat (softens within the range of 230°F–240°F)



Fig. 13.1: Gutta-percha sticks

Manipulation

Before insertion into the prepared cavity, the walls may be slightly moistened with solvents so as to soften gutta-percha and increase its adhesiveness. The material is carefully softened over an alcohol lamp or Bunsen burner. Avoid overheating as it could burn, leading to oxidation of its components, compromising with properties. It is inserted in cavity in bulk and condensed with wet smooth burnisher of suitable size. Remove excess material immediately with a hot/flat spatula. Smoothen the surface with slightly warm instrument. Ask the patient to bite and check for occlusion.

Physical properties

- Almost colorless with a slight pinkish or greyish hue.
- Odorless.
- Slightly elastic and contracts on cooling.
- Non-irritating to the soft tissues.
- Softens on heating and hardens again on cooling.
- Readily soluble in chloroform and partially soluble in eucalyptus oil.
- Specific gravity is 0.96–1.0.
- Low crushing strength and abrasion resistance.
- Ease of insertion and removal, when desired.

Disadvantages

- Strength and abrasion resistance are poor.
- Excessive shrinkage on cooling with consequent tendency to draw away from the cavity walls during hardening.
- When exposed to air, gutta-percha deteriorates to a hard brittle material because of oxidation.
- Becomes porous and disintegrates after some time.
- Very high coefficient of thermal expansion (198 × 10^{-6}/°C).
- Higher plastic flow.

II. Dental Cements

Cements are the most widely used materials for temporary restorative purposes. The cements used as interim restorations are summarized in Table 13.1.

Table 13.1: Cements employed for interim restorations

Cement	*Primary use*	*Secondary use*	*Pulp response*	*Anticariogenic potential*
Zinc oxide eugenol	• Interim restoration • Luting agent	• Periodontal bandage • Root canal obturation	Mild	No
Cavit	• Interim restoration • Insulating base	• Interim restoration • Insulating base	Nil	Nil
Zinc phosphate	• Luting agent	• Interim restoration • Thermal insulating base	Moderate	No
Polycarboxylate	• Luting agent • Insulating base	• Luting agent • Interim restoration	Mild	Yes
Calcium hydroxide	• Thermal base • Pulp capping agent	• Interim restorations • Liner	Soothing	No

The commonly used materials are as follows.

a. *Zinc Oxide Eugenol Cement and its Modified Forms*

Zinc oxide eugenol is the most widely used temporary restorative material.

Zinc oxide eugenol (Fig. 13.2) offers much better biocompatibility than rest of the dental cements. It is a good insulator and sealer of pulp dentin organ. Additionally, it has an antiseptic effect on micro-organisms remaining in dentin, and also has a sedative and anti-inflammatory action on pulp dentin organ. When inserted into the cavity, it has an approximately neutral pH of 7.0. Unfortunately, its strength is not sufficient enough to resist forces of mastication, lacks resistance to wear and has a relatively high solubility in the oral cavity.

Basically, the types of zinc oxide eugenol cements available are:

i. Type I (used as a base or temporary restoration)
ii. Type II (used as a cavity liner and cementation of appliances).

> *Zinc oxide eugenol cements differ from zinc oxide eugenol impression pastes in that the latter contains plasticizer, like linseed oil, olive oil, mineral oil, etc., which increases its flow.*

Early zinc oxide eugenol cement was easy to mix and place; however, it was hydrolytically unstable and lacked adequate strength.

To improve upon the working, handling and setting characteristics, various modifications have been tried; mainly following two approaches:

i. Partial substitution of eugenol by o-ethoxybenzoic acid (EBA)
ii. Addition of fused quartz/aluminum oxide/resin polymer to the powder component

The modified forms of zinc oxide eugenol are:

i. *Resin-reinforced zinc oxide eugenol cements:* 20% polymethyl methacrylate is usually added to zinc oxide as powder and is mixed with eugenol along



Fig. 13.2: Zinc oxide eugenol



Fig. 13.3: Resin-modified zinc oxide eugenol

with accelerators, like zinc acetate and zinc stearate (Fig. 13.3). Thymol and hydroxyquinoline as antimicrobial agents are added in traces.

Composition	
Powder	*Weight %*
Zinc oxide	80.0%
Polymethyl-methacrylate	20.0%
Zinc stearate	Traces
Zinc acetate	Traces
Thymol	Traces
Liquid	
Eugenol	85%
Olive oil	15%

Resin improves the strength, homogeneity and smoothness of the mix, decreases flow, brittleness and solubility. Resin may physically bond to the other components or react chemically with the eugenol. Zinc stearate acts as a plasticizer and zinc acetate improves the strength of the cement. Zinc stearate and zinc acetate also act as accelerators. Olive oil acts as a plasticizer and masks the irritating effects of eugenol.

ii. *EBA (orthoethoxybenzoic acid) modified cement:* Orthoethoxybenzoic acid as chelate was tried to improve upon the physical properties of zinc oxide eugenol.

Composition	
Powder	
Zinc oxide	70%
Alumina/fused quartz/calcium phosphate	30%
Liquid	
EBA	62.5%
Eugenol	37.5%

Addition of EBA to eugenol resulted in the following properties:

- Increases the amount of powder that can be incorporated to liquid to obtain a mixture of standard consistency.

- Increases compressive, tensile and shear strength (closely resembles that of zinc phosphate).
- Decreases the setting time, only if EBA concentration is less than 70%. More than 70% EBA increases the setting time sharply.
- The density increases from 2.68 g/mL for conventional zinc oxide eugenol to 3.31 g/mL for EBA cement.
- Increases solubility and disintegration characteristics of the products from 0.04% for zinc oxide eugenol to 0.05% for EBA cement.
- The linear coefficient of thermal expansion increased to 60×10^{-6}/°C as compared to 35×10^{-6}/°C for zinc oxide eugenol.
- Addition of EBA does not have any adverse effect on pulp.

Manipulation

Zinc oxide eugenol is usually available in the powder and liquid form. Mixing can be carried out with a stainless steel spatula over a clean glass slab. Disposable paper pads can also be used. The required amount of powder and liquid as instructed are dispensed onto the slab. The powder in bulk is incorporated into the liquid and thoroughly spatulated. A series of smaller amounts are added subsequently and spatulated. The correct consistency is achieved when the mix is heavy putty like and can be picked up in the fingers without sticking. The heavy mix of cement is rolled into a rope and kept ready for use.

With the plastic instrument, the cavity is filled and the excess is removed carefully. A large cotton pellet saturated with hot water is patted over the entire surface of the cement. The hot water hastens the setting of the cement, imparting harder surface.

Effects on pulp

Zinc oxide eugenol, as interim material, prevents leakage and subsequent growth of bacteria. The beneficial effects on pulp are:

i. *Bactericidal effect:* Eugenol is bactericidal at relatively high concentrations of 10^2–10^3 mol/L (1 mole is equal to molecular weight of the compound; mol/mole is the concentration of that substance in one litre). Bactericidal effect is beneficial; however, higher concentrations are toxic to pulp.

ii. *Sedative effect:* At low concentrations, eugenol acts like a local anesthetic. It decreases intradentinal fluid activity minimizing the sensitivity to hot, cold or sweet. Eugenol also inhibits neural conduction.

iii. *Anti-inflammatory effect:* Eugenol in low doses causes resolution of mild inflammation. It inhibits neutrophil function and chemotaxis, and removes harmful free radicals, like superoxide, which are generated by these cells.

b. *Cavit*

Cavit is a premixed non-eugenol paste used for temporary restorations and cavity bases (Fig. 13.4). It contains zinc oxide, zinc sulfate, calcium sulfate, glycol acetate, polyvinyl acetate, polyvinyl chloride acetate, triethanolamine and red pigments.

When inserted into dry cavity preparations, cavit creates a negative pressure, causing aspiration of odontoblasts leading to pain. However, no aspiration is evident when the cavity is moistened with water or eugenol.

Cavit is not a satisfactory material for cementation because of its film thickness (91 μ) and hygroscopic expansion. This causes the restoration to rise producing hyperocclusion. Compressive strength of Cavit is approximately 2000 psi, half that of zinc oxide eugenol. pH of both the materials is almost the same, i.e. 6.9–7.0.

A variety of non-eugenol zinc oxide cements are available in market (Figs 13.5 and 13.6).



Fig. 13.4: Cavit (non-eugenol interim restorative material)



Fig. 13.5: Cimpat blanc (non-eugenol interim restorative material)



Fig. 13.6: Litark (non-eugenol interim restorative)



Fig. 13.7: Zinc phosphate cement (powder and liquid)

c. Zinc Phosphate Cement and its Modified Forms

Zinc phosphate cement (Fig. 13.7) is of two types depending upon the particle size:

Type I (fine grain, film thickness 25 μ or less; designed for seating precision appliances).

Composition	
Powder	*Weight %*
Zinc oxide	90.2%
Magnesium oxide	8.2%
Silicone dioxide	1.4%
Bismuth trioxide	0.1%
Barium oxide	Traces
Barium sulfate	Traces
Calcium oxide	Traces
Liquid	
Phosphoric acid (free acid)	38.2%
Phosphoric acid (combined with aluminum and zinc)	16.2%
Aluminum	2.5%
Zinc	7.1%
Water	36.0%

Type II (medium particle size, film thickness 40 μ; used as interim restorations and thermal insulating bases).

Manipulation

A proper amount of powder is to be incorporated into the liquid to achieve better properties of the cement. The features involved in manipulation are:

- *Mixing slab:* Since combination of powder and liquid is a highly exothermic reaction, it is necessary to dissipate the heat, otherwise the reaction will proceed too rapidly towards completion. A properly cooled glass slab (18–20°C) is used to dissipate heat. The temperature of the slab should not be lowered around the dew point, otherwise water droplets will form and contaminate the mix.
- *Powder/liquid ratio:* A larger powder/liquid ratio greatly improves the properties of the mixed cement; powder/liquid ratio used for interim restorations is 2.5:1.
- *Care of the liquid:* The liquid of the cement should always be stored in stoppered bottles and exposure to air should be avoided as far as possible. Exposure to air will lead to change in the water/acid ratio, which affects the subsequent setting and working qualities of the cement. The liquid exposed to humid environment, absorbs water from the surroundings; whereas when exposed to dry environment, it loses water. When the water content of the liquid is reduced, it is often evident by the formation of crystals on the walls of the bottle, or a general cloudiness of the liquid. Unfortunately, if water is absorbed by the liquid, no change in appearance can be observed. Repeated opening of the bottle over a long period of time undoubtedly alters the water/acid ratio of the remaining liquid. Therefore, approximately the last one-fifth of the liquid should be discarded.

Mixing is accomplished with a long narrow bladed stainless steel spatula on a cool glass slab. The desired amount of powder is placed on the slab and divided into 6–7 increments. First the smallest increment is incorporated and spatulated over a large area for approximately 15 seconds followed by rest of the increments one by one. The cement is mixed with the spatula held flat on the surface of the slab, moved in a rotary motion.

Mixing over a large surface area on a cooled glass slab allows the heat to be dissipated most effectively. Incorporation of small increments at the start allows partial neutralization of the liquid; hence, the temperature rise would not be high. Later on, larger amounts of powder incorporated during the middle of the mixing period, saturates the liquid with the newly forming zinc phosphate. The quantity of unreacted acid is less at this time because of the prior

neutralization of the acid. The amount of heat liberated is also less. Finally, smaller increments are added, so that the desired consistency is achieved. Thus the mixing procedures begin and end with smaller increments, first to obtain slow maturation of the liquid and finally to gain a critical consistency.

The pH of zinc phosphate cement is approximately 3.5 in the beginning. The pH increases rapidly approaching neutrality in 24–48 hrs. Therefore, any damage to pulp from acid attack by zinc phosphate cement probably occurs during the first few hours after insertion. Pulp protective measures are required, especially in deep cavities.

Setting time

The setting time is 4–7 minutes. Several factors influence the rate of setting of zinc phosphate cement, such as:

i. *Under control of the manufacturer:*
   - Composition and sintering temperature of the powder.
   - Composition of the liquid (water content and amount of buffering salts).
   - Particle size of the powder (smaller the particle size, greater the surface area in contact with the liquid and hence, rapid is the reaction).

ii. *Under control of the operator:*
   - *Powder/liquid ratio:* Working and setting times can be increased, by decreasing the powder/liquid ratio. (The procedure is not acceptable as reduction in powder/liquid ratio leads to decrease in strength and pH of the cement.)
   - *Rate of powder incorporation:* Introduction of a small quantity of powder into the liquid increases the setting time by reducing the amount of heat generated, especially for first few increments.
   - *Mixing temperature:* Decreasing the temperature by cooling the glass slab increases the setting time.
   - *Spatulation time:* Prolonging the spatulation time destroys the forming matrix. Fragmentation of the matrix needs extra time to rebuild the bulk of matrix.
   - *Water contamination or loss of water:* The addition of water decreases setting time and loss of water increases setting time.

Modified zinc phosphate cements

i. *Fluoridated cements:* 10% stannous fluoride is added to zinc phosphate powder. Fluoride release continues resulting in reduced enamel solubility and increased hardness. These cements have low strength and high solubility than conventional zinc phosphate cements.

ii. *Copper cements:* Copper cements may consist of a portion of cuprous oxide (red) or cupric oxide (black) added to the zinc oxide powder. The main object of addition of copper is to render the cements germicidal in the oral cavity. There is hardly any advantage of these cements over conventional cements.

d. *Polycarboxylate Cements*

Polycarboxylate cements have the strength and manipulative properties comparable to phosphate cements (Fig. 13.8). The advantage of this cement is its adhesive qualities, low irritation potential and also the anticariogenic properties.

Composition
Powder
• Modified zinc oxide • Stannic oxide may be substituted for magnesium oxide. • Bismuth and aluminum oxide • Small quantities of stannous fluoride.
Liquid
• Aqueous solution of polyacrylic acid • Itaconic/tartaric acid.

Manipulation

The recommended powder/liquid ratio for interim restorations is 2:1. Polycarboxylate cement, especially the liquid should not be stored in a refrigerator; the low temperature may lead to gel formation in liquid.

Mixing is carried out on a cool glass slab. The working time ranges from 3–5 minutes.

The liquid should be dispensed just before the mixing is to be started, because exposure of the liquid to air even for a few minutes results in evaporation of the water; subsequently, increasing the viscosity of the liquid.

Powder is incorporated into the liquid in two or three increments, using rapid spatulation. The mix should be completed in 30 seconds to allow maximum working time.



Fig. 13.8: Polycarboxylate cement (powder and liquid)

Polycarboxylate cements have thixotropic properties; the viscosity decreases when shear forces are applied as during spatulation or when placing a restoration.

The larger size of the polyacrylic acid molecule and its binding to calcium and proteins limits its diffusion through the dentinal tubules. Polyacrylic acid is a weak acid and causes minimal movement of the fluid in the dentinal tubules. Additionally, it is biocompatible with the pulp. All these factors make polycarboxylate cement less irritating than zinc phosphate cements.

e. Calcium Hydroxide

Being compatibility with composites, calcium hydroxide is used as a base and also as an interim restoration. Dycal, the commonly available calcium hydroxide preparation consists of two tubes; one containing base and the other catalyst (Fig. 13.9).

Composition (Dycal)
Base
• Zinc oxide • Calcium phosphate • Calcium tungstate/Barium sulphate • Iron oxide/titanium oxide • Glycol salicylate
Catalyst
• Calcium hydroxide (50%) • Zinc oxide (10%) • Zinc stearate (0.5%) • N-ethyl p-toluene sulfonamide (39.5%)

The base and the catalyst are mixed in the ratio of 1:1 on a clean disposable pad. The setting time is approximately one minute outside the oral cavity. However, it decreases substantially inside the oral cavity because of higher temperature and humidity.



Fig. 13.9: Quick setting calcium hydroxide (Dycal)

Light cure calcium hydroxide is also used as interim material. These materials exhibit better compressive and tensile strengths and less solubility in oral fluids. The calcium hydroxide preparation is preferred as an interim restoration, in case where composite is to be used as final restoration. Half of the filling can be removed and the other half can be used as a base under the composite restoration. Calcium hydroxide is very effective in cases of rampant caries. The progress of rampant caries is very quick, so immediate permanent restoration is not preferred. Progress/arrest of caries can be evaluated before final restoration is given.

The physical properties of cements used as interim restorations are summarized in Table 13.2.

B. FOR EXTRACORONAL PREPARATION

I. Prefabricated Crowns

Various types of prefabricated crowns are available, such as aluminum, stainless steel, celluloid and polycarbonate crowns (Fig. 13.10). Crown forms available in anatomic shapes are most effective because of the function and protection they offer to the gingival tissues. Transparent crowns are usually used for anterior teeth and the metal crowns for the posterior teeth. If anatomic crowns are not available, these can be cut according to the contour of the tooth to be restored. The cut edges are smoothened with a mounted stone or rubber tips. Usually, these crowns are kept slightly away from the gingival margins, since these may lead to gingival recession at those areas. The patient is instructed to brush carefully so as not to injure the gingival margin.

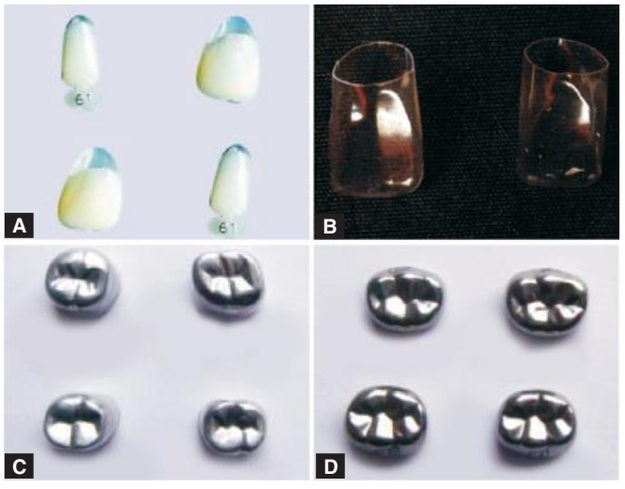


Fig. 13.10: Prefabricated interim restorations: **(A)** Polycarbonate crowns; **(B)** Cellulose crowns; **(C)** Aluminum crowns; **(D)** Stainless steel crowns

Table 13.2: Physical properties of cements used for interim restoration

Cement	*Composition*		*Powder: liquid ratio*	*Compressive strength psi (MPa)*	*Tensile strength psi (MPa)*	*Solubility in H_2O (% wt)*	*Film thickness (μm)*	*Modulus of elasticity (psi)*	*Setting time (minutes)*	*pH at 24 hrs*	*Coeff. of thermal expansion*
	Powder	*Liquid*									
Zinc oxide eugenol	Zinc oxide 70% Rosin 30% Zinc acetate traces	Eugenol 100%	3:1	800–4000 (6–28)	300 (2.1)	0.04	25–30	0.03	24 hrs	7.0	35×10^{-6}/°C
EBA-alumina reinforced ZnOE	Zinc oxide 70% Alumina 30%	Ethoxybenzoic acid 62.5% Eugenol 37.5%	4:1	8700–11600 (60–80)	800 (5.8)	0.05	30–40	0.79	7–9	6.7	66×10^{-6}/°C
Polymer reinforced ZnOE	Zinc oxide 80% Polymethyl 20% methacrylate	Eugenol 85% Olive oil 15%	4:1	5400–5800 (37–40) (4.1)	600	0.08	30–32	0.39	6–9	6.8	60×10^{-6}/°C
Zinc phosphate	Zinc oxide 90.2% Magnesium oxide 8.2%, silica dioxide 1.4%, bismuth dioxide 0.1%, misc. 0.1%	Phosphoric acid (54.5%) Water 36% Others 0.6%	2.5:1 for filling 1.5:1 luting	15000 (104)	800 (5.5)	0.1–0.3	Type I -25 Type II-40	1.96	4–7	Initially 3.5; after 24 hours, 5.5	—
Polycarboxylate	Zinc oxide 90% Magnesium oxide 10%, stannous fluoride (traces) or alumina 30% replaces zinc oxide	40% polyacrylic acid Itaconic acid water	2:1 for filling and 1:1 for luting	8300–14000 (57–99)	900 (6.2)	0.1–0.25	25–30	0.74	7–9	Initially 1.5, soon = 5.5	—
Zinc silicophosphate	Silica 35% Alumina 25% Zinc oxide 10–20% Fluoride 17% Calcium oxide 1–50%	Phosphoric acid 48% Water 35% Buffers 17%	2.5:1 for filling and 1.5:1 for luting	21000 (145)	1100 (7.6)	0.4	30	—	3.5–4	Initially 1.5, At 24 hours = 5.0	—
Resins	Resin matrix, inorganic fillers, organo silanes, photo or chemical initiators and activators	Methyl methacrylate, tertiary amine		10000–25000 (70–172)	—	0–0.01	>25	0.31–0.46	2–4	—	20–25 × 10^{-6}/°C
Calcium hydroxide				10–27	—	0.4–7.8	—	0.37	2.5–5.5	10–12.3	—

The crowns are filled with a thin mix of zinc oxide eugenol cement and fixed onto the tooth. The excess cement is wiped out. The patient is asked to bite smoothly and the occlusion is checked. 3.0% hydrogen peroxide can be used to clean the marginal area of remaining cement.

II. Indirect Acrylic Restorations

Several resins are being used to make temporary restorations; however, self-curing acrylic is the widely used material (Fig. 13.11). Acrylics can be used both directly over the tooth or fabricated outside the oral cavity and then seated. The major drawback of using direct acrylic is that the free monomer and the heat produced during exothermic reaction are harmful to the pulp and the gingival tissues. The disadvantage of indirect method is the time required for fabrication of the restoration. However, the material provides accuracy, appearance and function as required for interim restorations.

Technique

A preoperative rubber base impression of the arch is taken and stored. In case the tooth to be restored is partially broken or otherwise mutilated, the cast is prepared from the impression and the tooth is build with wax over the cast. A fresh impression is taken from that cast and stored.

The tooth preparation is completed. After gingival retraction, a rubber base impression is taken and a cast is made. The tooth to be restored is coated with cold mold seal. Self-cure acrylic resin of the desired shade is mixed in a dappen dish and poured into the impression already stored. The cast is inserted into the impression and kept there for a few minutes. The cast and the impression are separated after some time. A temporary crown is ready simulating shape, size and contour of the original tooth. The same is finished, polished and seated onto the prepared tooth. Zinc oxide eugenol should not be used, if the temporary crown is to be used for a longer time, since eugenol weakens the acrylics. The gingival area is cleaned and the occlusion is finally checked.



Fig. 13.11: Self-cure acrylic resin—powder and liquid

Indirect acrylic restorations can also be fabricated over the cast, but such restorations may not match the configuration of original tooth.

Interim restorations are mandatory for the protection of pulp and gingival tissues. Along with sealing the cavity preparation, they also provide esthetics and partial functioning of teeth. For extracoronal preparation, they stabilize the tooth movement till permanent restoration is fabricated. The interim restorations add boon to the clinical success.

BIBLIOGRAPHY

1. Akova A, Ozkomur, Uysal H. Effect of food-simulating liquids on the mechanical properties of provisional restorative materials. Dent Mater 2006;22:1130–34.
2. Burke FJ, Murray MC, Shortall AC. Trends in indirect dentistry provisional restorations. Dent Update 2005; 32:443–47.
3. Gilles JA, Huget EF, Stone RC. Dimensional stability of temporary restorative materials. O Surg, O Med, O Path 1975;40:796–99.
4. Gough M. A review of temporary crown and bridges. Dent Update 1994;16:203–07.
5. Gratton DG, Aquilino SA. Interim restorations. Dent Cl North Am 2004;48:487–502.
6. Hammod BD, Cooper JR, Lazarchik DA. Predictable repair of provisional restorations. J Esthet Restor Dent 2009;2:19–23.
7. Kerby RE, Knobloch LA, Sharples S, Peregrina A. Mechanical properties of urethane and bis-acryl interim resin materials. J Prosthet Dent 2013;110:21–28.
8. Lui JL, Setcos JC, Phillips RW. Temporary restorations: a review. Oper Dent 1986;11:103–07.
9. Nejactidanesh F, Lotfi HR, Savabi O. Marginal accuracy of interim restorations fabricated from four autopolymerizing resins. J Prosth Dent 2006;95:364–68.
10. Nemkovsky CE, Gross MD. Transferring provisional restorations to final master casts. J Oral Rehab 1994; 21:157.
11. Patras M, Naka O, Duokoudakis S, Pissiotis A. Management of provisional restorations' deficiencies: a literature review. J Esthet Restor Dent 2012;24:26–38.
12. Prashad K, Shetty M, Alva H, Prasad A. Provisional restorations in prosthodontic rehabilitations-concepts, materials and techniques. Nitte Univ J Health Sci 2012; 2:72–77.
13. Rakhshan V. Marginal integrity of provision resin restoration materials: A review of the literature. Saudi J Dent Res 2015;6:33–40.

14. Regish KM, Sharma D, Prithviraj DR. Techniques of fabrication of provisional restoration: An overview. Int J Dent 2011;Art. ID134659.
15. Rutkunas V, Sabaliauskas V, Mizutani H. Effects of different food colorants and polishing techniques on color stability of provisional prosthetic materials. Dent Mater J 2010;29:167–76.
16. Schulz SD, Laquai T, Kummerer K, Bolek R, Mersch-Sundermann V, Polydorou O. Elution of monomers from provisional composite materials. Int J Polymer Sci 2015; 1–7.
17. Schwartz JK, Oliveira-Ogliari A, Meereis CT, Leal FB, Ogliari FA, Moraes RR. Characterization of Bis-Acryl composite resins for provisional restorations. Braz Dent J 017;28:354–61.
18. Small BW. Indirect provisional restorations. Gen Dent 1999;47:140–42.
19. Usumez A, Ozturk AN, Aykent F. The effect of dentin desensitizers on thermal changes in the pulp chamber during fabrication of provisional restoration. J Oral Rehab 2004;31:579.
20. Vaidyanathan T, Vaidyanathan J, Manasse M. Analysis of stress relaxation in temporization materials in dentistry. Dent Mater 2015;31:e55–e62.
21. Waerhaug J. Temporary restorations: Advantages and disadvantages. Dent Clin North Am 1980;24:305–16.
22. Walter RD, Goodacre CJ. The esthetic and psychologic benefits of an intraoperative provisional restoration. J Esthet Restor Dent 2017;29:189–92.
23. Wang RL, Moore BK, Goodacre CJ, Swartz ML, Andres CJ. A comparison of resins for fabricating provisional fixed restorations. The Int J Prosthodont 1989;2:173–84.
24. Wassell RW, St George G, Ingledew RP, Steele JG. Crowns and other extracoronal restorations: provisional restorations. Br Dent J 2002;192:619–23.
25. Xie D, Fadah M, Park JG. Novel amino acid modified zinc polycarboxylates for improved dental cements. Dent Mater 2005;21:739–45.
26. Yao J, Li J, Wang Y, Huang H. Comparison of the flexural strength and marginal accuracy of traditional and CAD/CAM interim materials before and after thermal cycling. J Prosthet Dent 2014;112:649–57.
27. Youdelis RA, Faucher R. Provisional restorations: An integrated approach to periodontics and restorative dentistry. Dent Clin North Am 1980;24:285–303.